FUNCTIONAL STRUCTURE(S), FORM AND INTERPRETATION

Perspectives from East Asian languages

Edited by Yen-hui Audrey Li and Andrew Simpson

RoutledgeCurzon
Taylor & Francis Group
LONDON AND NEW YORK

Errata

p. 34, (27):	recommend-ever	→	recommend-even	
p. 41, 2nd Paragraph:	*so* in Japanese	→	*so*- in Japanese	
p. 55, just before (96):	be analyzes	→	be analyzed	(twice)
p. 57, just before (102):	*to* and	→	-*to* and	
p. 58, (104 b-c):	], $_{\text{and}}$	→	], and	
p. 60, footnote 4, (i):	**Kosunaa**-ga	→	**Kosunaa$_{1}$**-ga	

2

COPYING VARIABLES*

Yoshihisa Kitagawa

1. Introduction

It is well known that ellipsis involving a proform as its part can exhibit flexibility in anaphoric interpretations – often in somewhat unexpected ways. In this work, I will attempt to show that when we clarify how anaphora involved as part of ellipsis comes to be represented in covert syntax, we will also have a better understanding of the way the so-called E-type anaphora (including donkey anaphora) is represented in covert syntax. In particular, I will first point out that ellipsis can provide a type of strict identity interpretation for a 'reconstructed' proform which is akin to the interpretation recognized in the E-type anaphora. I will then argue that the parallelism between the two constructions arises due to the involvement of the same operation of 'reconstruction' in the form of copying applying in covert syntax. It should be made clear at this point that the research presented in this work deals mostly with the syntactic aspects of these phenomena and leaves out their semantics. We will, in other words, attempt to answer the question what syntactic operations are responsible for the semantic characteristics of these constructions and how they should be represented at LF, but will leave unanswered the question how they should be represented semantically. In the final part, I will extend the proposed analysis to donkey anaphora, critically examining some alternative approaches.

The theoretical framework I will adopt in this work is one version of the minimalist program. I will, for instance, follow Chomsky (1995) and assume that grammar is constrained by various types of 'minimalism' imposed by the Bare Output Conditions and certain economy conditions. Clearly departing from the standard minimalist assumption, however, I will attempt to advocate an approach in which covert syntax can be driven by factors other than formal feature checking.[1] I will hypothesize, in particular, that covert syntax can be triggered by any factor that achieves legitimacy of syntactic objects at LF.

Implementing this working hypothesis, we can, for instance, offer a very simple account of the well-known ambiguity observed in VP-Ellipsis in (1).

(1) John loves his wife, and Bill does $[_{VP}$ e], too.

In particular, the two distinct interpretations of the second clause – 'Bill loves his own wife' (sloppy identity) and 'Bill loves John's wife' (strict identity) – can be captured in terms of the two distinct orders in which 'linking' (for syntactic binding) and 'copying' (for the reconstruction of the elided VP) can apply as covert syntactic operations (Kitagawa (1991b), cf. Williams (1977)). For instance, as illustrated by the derivation in (2) below, sloppy identity arises when nominal binding applies **after** the reconstruction of the VP takes place.

(2) a. LF$_j$: John$_1$ [$_{VP}$ loves his wife], and Bill does **[$_{VP}$ love *his* wife]**, too.

b. LF$_j$: *John*$_1$ loves ***his*** wife, and *Bill*$_2$ does [$_{VP}$ love ***his*** wife], too.

Sloppy identity arises in (1), in other words, when the VP containing an **unbound** proform is reconstructed at the ellipsis site. Strict identity arises, on the other hand, when the VP containing a **bound** proform is reconstructed. That is, when syntactic binding applies before VP-reconstruction applies, as illustrated by the derivation in (3).[2]

(3) a. LF$_i$: *John*$_1$ [$_{VP}$ loves ***his*** wife], and Bill does [$_{VP}$ e], too.

b. LF$_j$: John$_1$ [$_{VP}$ loves his$_1$ wife], and Bill does **[$_{VP}$ love *his*$_1$ wife]**, too.

A virtually identical analysis permits us to capture the sloppy-strict ambiguity observed in the empty nominal (or DP-Ellipsis) construction in Japanese as in (4).[3]

(4) *John-wa* [$_{DP}$ **zibun**-*no ansyoo-bangoo*]-*o wasuretesimatteita ga,*
-TOP self-GEN PIN.number-ACC forgot though
Okusan-wa **[$_{DP}$ e]** *oboeteita.*
Wife-TOP remembered
'While John forgot his PIN number, his wife remembered { his / her } PIN number.'

Here, the elided DP can be interpreted either as 'John's wife's own PIN number' (sloppy identity) or 'John's PIN number' (strict identity), and each of these readings can be ascribed to the distinct order in which the two covert operations applied. When the reconstruction of DP takes place first, and then the syntactic binding of *zibun* 'self' does as in (5) below, an **unbound** proform is reconstructed at the ellipsis site, and sloppy identity arises.

(5) a. LF$_{i*}$: *John-wa* [$_{DP}$ *zibun-no PIN*]-*o* ... *Okusan-wa* **[$_{DP}$ zibun-no PIN]**-*o* ...

b. LF$_k$: *John*$_1$-*wa* **zibun**$_1$-*no PIN-o* ... *Okusan*$_2$-*wa* **zibun**$_2$-*no PIN-o* ...

When the order of application is reversed as in (6) below, on the other hand, a **bound** proform is reconstructed at the ellipsis site, and strict identity arises.[4]

(6) a. LF_i: *John*$_1$*-wa* [$_{DP}$ **zibun**$_1$*-no PIN*]*-o* ... *Okusan*$_2$*-wa* **[$_{DP}$ e]** *oboete...*

b. LF_k: *John*$_1$*-wa* [$_{DP}$ *zibun*$_1$*-no PIN*]*-o* ... *Okusan*$_2$*-wa* **[$_{DP}$ zibun$_1$-no PIN]***-o* ...

Finally, this analysis allows us to assimilate certain interpretations of overt pronouns to those involved in the ellipsis constructions we have examined above. The sloppy identity of a 'pronoun of laziness' in the celebrated 'paycheck sentence' as in (7) below, for instance, arises when the reconstruction of the pronoun's DP antecedent is followed by the syntactic binding of the contained pronoun as in (8) (cf. Jacobson (1980)). (For ease of explanation, we relate *the man* and *his* directly, disregarding *who* and its trace.)

(7) The man who gave [$_{DP3}$ **his**$_1$ paycheck] to his wife is wiser than the man who gave **it**$_3$ to his mistress.

(8) a. LF_i: The man$_1$ who gave [$_{DP}$ his paycheck] to his wife is wiser

than the man who gave **[$_{DP}$ his paycheck]** to his mistress.

b. LF_j: *The man*$_1$ who gave [***his***$_1$ paycheck] to his wife is wiser

than *the man*$_2$ who gave [***his***$_2$ paycheck] to his mistress.

While the pragmatics does not permit strict identity in (7), a similar sentence as in (9) below exhibits strict identity, and this interpretation can be derived when the syntactic binding takes place before DP-reconstruction does, as illustrated in (10).

(9) The man who$_1$ had **[$_3$ his$_1$ car]** stolen suspects the teen-age boy who was staring at **it**$_3$ from a nearby truck when he left the parking lot.

(10) a. LF_i: *The man*$_1$ who had **[$_{DP}$ *his*$_1$ car]** stolen suspects the teen-age boy

who was staring at **it** ...

b. LF_j: The man$_1$ who had [$_{DP}$ *his*$_1$ car] stolen suspects the teen-age boy$_2$

who was staring at **[$_{DP}$ his$_1$ car]** ...

Kitagawa (1995) argues that a similar covert reconstruction is involved even in the coreference between a name and a pronoun as in (11) below(, in which no 'deictic' use, i.e., neither physical nor psychological pointing, is intended for the use of a pronoun.)

(11) a. LF_i: [***John***$_1$'s mother] saw ***him***$_1$. ⇒
b. LF_j: [*John*'s mother] saw ***John***

In the same work, it was pointed out that this 'LF-reconstruction' approach to coreference allows us to formally assimilate 'referential circularity' as in (12) and 'i-within-i' violation as in (13).

(12) *[$_j$ **His**$_i$ wife] admires [$_i$ **her**$_j$ husband]

(13) *[$_i$ **his**$_i$ friend]

Note that, in this approach, the coreference between *His$_i$* and [$_i$ *her husband*] in (12) results in a containment relation after reconstruction by LF-copy takes place as illustrated in (14).

(14) LF: [[***her*** **husband**](’s) wife] admires [her husband]

This representation eventually yields an infinitely regressive anaphoric relation, which seems to be precisely what is causing 'i-within-i' violation (cf. Higginbotham (1983: 404)). A working hypothesis underlying this covert reconstruction approach is that each and every nominal expression satisfies the Bare Output Conditions at LF only as a 'fully interpretable' contentful item like a referential expression, an operator or a bound variable. We then expect none of the anaphoric proforms to remain as they are but to undergo some covert syntactic process to turn themselves into legitimate objects at LF. When we try to extended this hypothesis to all types of proforms (with proper adjustments), we end up with postulating an approach which treats ellipsis and pronouns on a par.

Since this approach permits, for instance, a DP antecedent to be reconstructed onto an overt pronoun as in (8a), (10b), (11b) and (14), one may express a concern that it will eventually result in a clash between two lexical expressions. This concern, however, can be dismissed once we decide to: (i) adopt **covert** reconstruction (i.e., LF-copying) of the antecedent in all types of ellipsis constructions, and (ii) adopt the Bare Output Conditions of Chomsky (1995). First, covert reconstruction for ellipsis presupposes the postulation of phonetically empty proforms like *[$_{VP}$ e]* and *[$_{DP}$ e]*, which are anaphorically related to their antecedents. In a sense, then, ellipsis is regarded as nothing but a type of anaphora involving a phonetically empty proform. The only distinction between the DP-Ellipsis as in (4) and the anaphora involved in (7), (9), and (11) therefore is that the former involves a phonetically empty proform while the latter involves a lexical (i.e., phonetically non-empty) proform. Furthermore, the Bare Output Conditions, one of the main tenets of the minimalist program, require the phonetic/phonological features of lexical items to be non-existent at the LF-interface. As far as the covert syntax is concerned, in other words, there should not exist any reason to discriminate between the phonetically empty proforms involved in ellipsis and overt proforms like pronouns of laziness. In both cases, when an anaphoric relation is established between a proform and its **mutually non-c-commanding** antecedent (which perhaps presupposes agreement in one or more of their formal features like category, number, person and gender), the semantic features of the latter is reconstructed onto the former. Based upon such considerations, we can assimilate all of VP-Ellipsis in (1), DP-Ellipsis in (4) and pronouns in (7), (9) and (11), rejecting the view that ellipsis involves full reconstruction at LF while anaphora does not. Note that postulating PF-deletion for ellipsis does not permit us to assimilate pronouns in (7) and (9) to ellipsis. (We will discuss PF-deletion further below.) This provides the general background for the analysis proposed below.[5]

2. Strict identity in ellipsis

Let us now examine the interpretation of VP-Ellipsis in English as in (15) and (16).

(15) [A statement made by the principal of a boys' school]
In our school, **every student**$_x$ [$_{VP}$ respects **his**$_x$ teacher], and the parents also expect me to [$_{VP}$ e].

(16) In this school, **every student**$_x$ [$_{VP}$ respects **his**$_x$ teacher], but unfortunately, the principal doesn't [$_{VP}$ e].

In both these sentences, the first clause involves a proform that is bound by a quantificational antecedent and is interpreted as a variable. When the VP in this clause is interpreted at the ellipsis site, the bound proform *his*$_x$ contained in the VP exhibits a type of 'strict identity,' which includes an interpretation akin to that of the anaphorically used demonstrative *that* and/or *those*. Thus, the elided VP in (15) is interpreted roughly as 'respect **that/those** teacher(s).'[6]

Similar interpretations are obtained also when the antecedent VP contains a bound anaphor *each other* as a possessor as in (17) below, provided that the speaker permits 'couple-internal' reciprocality in its antecedent clause as in (18) to begin with.

(17) [A statement made by a marriage counselor]
Every couple$_x$ [$_{VP}$ criticizes **each other**$_x$'s odd habits], and quite often, I am also inclined to [$_{VP}$ e].

(18) **Every couple**$_x$ criticizes **each other**$_x$'s odd habits.

In (15)–(17), ellipsis follows either an infinitival marker *to* or a negated auxiliary verb. It therefore seems inappropriate to reduce the strict identity in (16), for instance, to the pragmatics-based interpretation of *do* as a main verb. The strict identity in question, in other words, seems to be made possible indeed by VP-Ellipsis rather than by the pragmatic control of 'deep anaphora' in the sense of Hankamer and Sag (1976)).[7]

Furthermore, when the bound proforms *their* and *each other* in (19a–b) below are reconstructed at the ellipsis sites, they are interpreted as 'about forty percent of the students' and 'about forty percent of couples,' respectively, rather than as 'students' or 'couples'.

(19) a. The survey indicates that, in our school, **about forty percent of the students$_x$**, quite mistakenly, [$_{VP}$ consider **their$_x$** SAT scores as satisfactory], but as principal, I must say I cannot [$_{VP}$ e].
b. [A statement made by a marriage counselor]
About forty percent of couples$_x$ [$_{VP}$ complain about **each other$_x$**'s appearance(s)] even when I can find absolutely no reason to [$_{VP}$ e].

The reconstructed proform in each of these examples, in other words, is intended to refer to the members of the set linguistically specified by the quantified antecedent of the 'original' proform in the antecedent VP rather than to refer to the entire set of 'students' or 'couples,' which, one might claim, is contextually made available.

Moreover, the elided VP in (20) below is interpreted as 'like **each** contestant's performance.'

(20) The contestants came up to the stage and performed one by one.
Every contestant$_x$ seems to have [$_{VP}$ liked **his$_x$** performance],
but *each time*, the judge apparently didn't [$_{VP}$ e].

The availability of this distributive interpretation at the ellipsis site also suggests that the strict identity here is based upon the interpretation of the proform bound by a quantificational element, and that a contextually available set reading as mentioned above is irrelevant. In short, the strict identity in question does **not** seem to arise because of the reference to the pragmatic context but rather due to some syntactically established anaphoric relation.[8]

Returning now to the sentences in (15)–(17), let us provide the observation that their non-elliptical counterparts as in (21) and (22) below are not well-formed. In particular, the proform showing up in the second clause cannot be legitimately bound by its antecedent in the first clause.

(21) In our school, **every student$_x$** [$_{VP}$ respects his$_x$ teacher], but unfortunately, the principal doesn't *[$_{VP}$ respect *his$_x$ teacher]*.

(22) **Every couple$_x$** [$_{VP}$ criticizes **each other$_x$**'s odd habits], and quite often, I am also inclined to *[$_{VP}$ criticize ****each other$_x$****'s odd habits]*.

What this indicates is that when the proforms in (15)–(17) are interpreted at the ellipsis site, they **cannot** undergo the process of syntactic binding in that position. The unsuccessful binding in (21) and (22) also suggests, first, that the ellipsis in (15)–(17) cannot be handled by PF-deletion applying to (21) and (22) as their base-generated and hence LF representations. Second, these ellipsis constructions cannot involve simple and full reconstruction of the **unbound** proform by any of LF-Copy, semantic or pragmatic accommodation, or a version of the E-type strategy which regards proforms simply as descriptions in disguise.

One may try to account for the grammaticality of (15)–(17) by combining LF-Copy of the antecedent with the 'vehicle change' analysis proposed by Fiengo and May (1994). As illustrated in (23), for instance, vehicle change can yield a well-formed LF of (17) with *their* as the 'pronominal correlate' of (the trace of) *other's*.

(23) LF: **Every couple**$_x$ [$_{VP}$ criticizes **each other**$_x$'s odd habits], and quite often,
I am also inclined to *[$_{VP}$ criticize **their**$_x$ odd habits]*

Note that the locality constraint imposed on anaphor binding will be irrelevant at the ellipsis site in (23). A similar analysis applied to (24) below, however, would yield an incorrect result. Notice that vehicle change would incorrectly provide a well-formed LF as in (25) for the ungrammatical sentence (24).

(24) $^{?}$***Every couple**$_x$ [$_{VP}$ criticizes **each other**$_x$], and quite often,
I am also inclined to [$_{VP}$ e].

(25) LF: **Every couple**$_x$ [$_{VP}$ criticizes **each other**$_x$], and quite often,
I am also inclined to *[$_{VP}$ criticize **them**$_x$]*

We thus should not make an appeal to vehicle change to deal with the ellipsis in (16), either.[9]

Finally, suppose that one attempts to capture the strict identity in (15) by bringing the second clause containing the reconstructed proform into the domain of its antecedent in the first clause as illustrated in (26) below, along the line of 'dynamic binding' (Chierchia (1995), et al.). (The representation is simplified and the irrelevant details are omitted.)

(26) $\forall_x$ [$_{IP}$ student (x) respect $\wedge$ (x, x's teacher) $\wedge$ … respect (I, **x**'s teacher)]

Even if the reconstructed anaphor in (17) can be brought into the binding domain of *every couple* in a similar way, however, the subject of the second clause *I* (or *PRO* bound by *I*) would remain to be the local binder of the reconstructed anaphor *each other*, and the locality constraint on anaphor binding would still remain unsatisfied.[10] Moreover, quite importantly, it must be assumed that this strategy is not available when the proform to be 'dynamically bound' is base-generated as in (21). It therefore would remain unanswered why dynamic binding applies only when ellipsis is involved.

Let us now examine the interpretation of the empty nominal (i.e., DP-Ellipsis) in Japanese as in (27).

(27) *siritu-daigaku-no* **dono kyoozyu**$_x$*-ga* [$_{DP}$ **zibun**$_x$*-no gakusee*]*-o*
private-college-GEN which professor-NOM self-GEN student-ACC
*suisensite-***mo**, *Monbusyoo-wa* [$_{DP}$ e] *saiyoosi-nai-daroo.*
recommend-ever Ministry.of.Education-TOP employ-not-perhaps

'No matter which professor of a private college may recommend self's (= his or her own) student, the Ministry of Education will probably not employ those students.'

In this sentence as well, the first clause involves a proform bound by a quantificational antecedent, and the bound proform contained in the DP exhibits a type of 'strict identity' when this DP is interpreted at the ellipsis site. Thus, the elided DP in (27) is interpreted roughly on a par with *sono gakusee* '**that** student' or *sono kyoozyu-no gakusee* '**that** professor's student.' Here again, the non-elliptical counterpart of (27) as in (28) below does not permit a similar interpretation (with or without prosodic reduction of *zibun-no gakusee-o* 'self's student-ACC') since the proform *zibun* 'self' showing up in the second clause cannot be legitimately bound by its quantificational antecedent in the first clause.

(28) *siritu-daigaku-no dono kyoozyu$_x$-ga* [$_{DP}$ *zibun$_x$-no gakusee*]*-o*
suisensite-mo, Monbusyoo-wa [$_{DP}$ ***zibun**$_x$*-no gakusee*]*-o saiyoosi-nai-daroo*
*self-GEN student-ACC

This fact forces us to reject the analysis of (27) which lets PF-deletion of an DP apply in the base-generated representation like (28). Any reconstruction of the unbound *zibun* or its dynamic binding will be also inappropriate.

One may also try to account for the grammaticality of (27) by reconstructing only the head nominal from the first clause and derive a representation as in (29), for instance, as in Hoji's (1998) 'Supplied N head' analysis.

(29) *siritu-daigaku-no dono kyoozyu$_x$-ga* [$_{DP}$ *zibun$_x$-no gakusee*]*-o*
suisensite-mo, Monbusyoo-wa [$_{NP = N}$ **gakusee**]*-o saiyoosi*-nai-daroo.
student-ACC
'No matter which professor of a private college may recommend self's (= his or her own) student, the Ministry of Education will probably not employ **students**.'

Based upon the analysis of empty nominals as NPs rather than DPs, this approach attempts to reduce the strict identity in (27) to the definite interpretation a bare noun in Japanese can exhibit (among other interpretations), as exemplified by (30).

(30) *John-wa* **reezaa purintaa**-*o tukatta.*
-TOP laser.printer-ACC used
'John used **the** laser printer(s) / a laser printer / laser printers.'

Tomioka (1998), for instance, attempts to derive the definiteness associated with the interpretation of an empty nominal as in (27) by letting the 'iota' operation apply to the reconstructed nominal head as in (29). When the representation as in (29) contains a base-generated bare nominal *gakusee* 'student,' however, this nominal does not exhibit the interpretation comparable to the empty nominal in (27), but is interpreted as the generic '**student**.' The sentence therefore expresses that the Ministry of Education will probably not employ **any student** (as opposed to, for example, someone who has already been teaching as a full-time faculty member.) Anyone who attempts to ascribe the strict identity in (27) to the flexible interpretation of bare nominals in Japanese, therefore, would have to explain the absence of a similar interpretation in its base-generated counterpart.

Moreover, the sentence in (27) makes up a quite natural discourse when it is followed by a sentence like (31).

(31) *kokuritu-dai-no* *gakusee-zyanai-to-ne.*
national university-GEN student-must.be
'It's got to be a student of a **national university**.'

The students referred to by the empty nominal in (27), in other words, must be those from **private colleges** rather than just 'students.' The only plausible source of such a reading in (27), however, seems to be the anaphoric relation established by *zibun*, which in turn is contained in the DP *zibun-no gakusee* 'self's student,' the antecedent of the empty nominal. These facts suggest that the reconstructed content of the empty nominal in (27) is not just the head nominal but the entire DP.

To recapitulate so far, we have examined VP-Ellipsis in English as in (32) and (33) (= (15) and (17)) and DP-Ellipsis in Japanese as in (34) (= (27)), all of which can be descriptively characterized as in (35).

(32) [A statement made by the principal of a boys' school]
In our school, **every student**$_x$ [$_{VP}$ respects **his**$_x$ teacher], and the parents also expect me to [$_{VP}$ e].

(33) [A statement made by a marriage counselor]
Every couple$_x$ [$_{VP}$ criticizes **each other**$_x$'s odd habits], and quite often, I am also inclined to [$_{VP}$ e].

(34) *siritu-daigaku-no* **dono** **kyoozyu**$_x$-*ga* [$_{DP}$ **zibun**$_x$-*no* *gakusee*]-*o*
private-college-GEN which professor-NOM self-GEN student-ACC
suisensite-**mo**, *Monbusyoo-wa* [$_{DP}$ e] *saiyoosi-nai-daroo.*
recommend-ever Ministry.of.Education-TOP employ-not-perhaps
'No matter which professor of a private college may recommend self's (= his or her own) student, the Ministry of Education will probably not employ those students.'

(35) a. A proform in the first clause is **c-commanded** by its quantificational antecedent and is interpreted as a **bound variable**.
b. **Ellipsis** is involved in the second clause, and the ellipsis site is **outside** the **c-command** domain of its antecedent.
c. The proform in the antecedent clause is successfully interpreted also at the ellipsis site despite the **lack of a c-commanding binder**.
d. This **proform** is interpreted at the ellipsis site as a **definite** description of the members of the set defined by the quantificational antecedent of the original proform.

This definite description, moreover, can be quite comfortably paraphrased into an anaphorically used demonstrative expression like *that* or *those*.

Comparing (32)–(34) with their base-generated counterparts, we have also arrived at the interim conclusion that the derivation of these sentences does not involve any of PF-deletion, reconstruction of unbound proforms or head nominals, vehicle change, and dynamic binding.

This conclusion does not leave us too many options, but suggests, first, that the content of the elided phrase in each of (32)–(34) is **covertly reconstructed** from the antecedent clause, and second, that it is carried out **without** involving the reconstruction of the proform in its **unbound** state. I will therefore adopt the analysis in which the proforms in these sentences are **syntactically bound in the antecedent clause**, and then **covertly copied** into the ellipsis site.[11] Under this analysis, since the proform reconstructed at the ellipsis site in each of (32)–(34) has been already bound in the antecedent clause, it need not undergo the covert process of syntactic binding again. It thus escapes binding failure in the second clause of their base-generated counterparts, observed, for example, in (36) (= (21)).

(36) In our school, **every student$_x$** [$_{VP}$ respects his$_x$ teacher], but unfortunately, the principal doesn't *[$_{VP}$ respect *his$_x$ teacher]*, too.

This analysis is based upon the working hypothesis that a proform that has undergone syntactic binding maintains its bound status even after it is reconstructed elsewhere in the syntactic representation. This in fact is what we need to assume even for the strict identity involving non-quantificational antecedent of the proform, for example, for the strict identity of the reconstructed *each other* and *zibun* 'self' exhibit in (37) and (38) below, respectively. Recall that both *each other* and *zibun* must be bound to be well-formed at LF (See Footnote 4).

(37) **They$_1$** [$_{VP}$ liked **each other$_1$**'s papers], and I did [$_{VP}$ e], too.

(38) *John-wa* [$_{DP}$ **zibun**-*no* *ansyoo-bangoo*]-*o* *wasuretesimatteita* *ga*,
-TOP self-GEN PIN.number-ACC forgot though
Okusan-wa [$_{DP}$ e] *oboeteita*.
Wife-TOP remembered
'While John forgot his PIN number, his wife remembered { his / her } PIN number.'

Recall also that covert copying applies when a proform and its antecedent (and hence an 'elided' phrase and its antecedent as well) **do not c-command** each other.[12] Thus the proposed analysis allows us to capture all the descriptive characteristics of (32)–(34) described in (35) above except for (35d). It must be left unexplicated in this work how exactly the definiteness of the proform that is akin to the interpretation of an anaphorically used demonstrative expression arises at the ellipsis site in (32)–(34). Since the copying of a bound proform plays the central role in the proposed approach, however, this result is not at all surprising. Note that when we adopt a discourse constraint like the Novelty Condition (Heim (1982)), a copied bound proform should never be allowed to be reintroduced into the discourse as an indefinite item.

The LF-derivation of (16) is provided in (39).

(39) a. *Syntactic Binding*:
LF$_i$: In our school, **every student$_x$** respects **his$_{x\sigma}$** teacher,
but unfortunately, the principal doesn't [$_{VP}$ e]
b. *VP-Copy*:
LF$_j$: In our school, every student$_x$ [$_{VP}$ respects his$_{x\sigma}$ teacher],
but unfortunately, the principal doesn't *[$_{VP}$ respect **his$_{x\sigma}$** teacher]*

First, as in (39a), the syntactic binding of *his* takes place in the antecedent clause. Then, as in (39b), *his* as a bound proform (whose bound status is indicated by **a subscript-σ as a purely mnemonic marker rather than as part of a formal representation**) is copied into the ellipsis site as part of the copied antecedent **VP**.

The LF-derivation of the Japanese example (27) is also provided in (40) below, in which *zibun* as a bound proform is copied, this time as part of a copied **DP**-antecedent.

(40) a. *Syntactic Binding*:
LF$_i$: **siritu-daigaku-no dono kyoozyu$_x$**-*ga* [$_{DP}$ **zibun$_{x\sigma}$**-*no gakusee*]-*o* *suisensite* **mo,**
Monbusyoo-wa **[$_{DP}$ e]** *saiyoosi-nai-daroo.*
b. *DP-Copy*:
LF$_j$: *siritu-daigaku-no dono kyoozyu$_x$-ga* [$_{DP}$ *zibun$_{x\sigma}$-no gakusee*]-*o* suisensite mo,
Monbusyoo-wa [$_{DP}$ **zibun$_{x\sigma}$**-no gakusee]-*o saiyoosi-nai-daroo.*

3. From ellipsis to E-type anaphora

At this point, it is appropriate to bring E-type pronouns as in (41) into the scene.

(41) a. **Few congressmen** admire Kennedy(, and) **They** are very junior.
b. John owns **some sheep** and Harry vaccinates **them** in the Spring.
c. **A dog** came in. **It** lay down under the table.

E-type pronouns are known to exhibit the properties summarized in (42) (Evans (1977), Evans (1980)).

(42) a. They have **quantified** DPs as their **antecedents**.

b. They are **not c-commanded** by their antecedents.
c. Yet, they are successfully interpreted as **definite** descriptions.

We cannot help noting that the properties of E-type pronouns stated in (42) have much in common with those of the ellipsis constructions in (32)–(34) stated in (35) above. In particular, both phenomena permit an item involved in operator-variable binding to be successfully interpreted in a remote position outside the c-command domain of the involved operator. Also, definiteness arises in the resulting interpretation in both cases, yielding an interpretation akin to that of the anaphorically used demonstrative expression *that* or *those*. I therefore would like to propose and argue that E-type anaphora and the ellipsis constructions in (32)–(34) are almost completely parallel – the syntactic derivation of both phenomena involves identical mechanisms, and they both yield the same type of strict identity, involving the copying of the projection of the category D(eterminer) in its bound state.

Near complete assimilation of the two phenomena can be achieved when we pay attention to the fact that they both involve a quantified element in the antecedent clause, which necessarily induces an operator-variable relation. The first thing we must do is to clarify how such a relation comes to be represented in covert syntax. Let us begin with the raising of the external argument under the Internal Subject Hypothesis. The first sentence in (41) above, for example, will be represented as in (43a) below after this operation takes place.

(43) a. *Subject Raising*:
Few congressmen$_y$ $[_{VP}$ **[e]**$_y$ admire Kennedy].

b. *QR*:
LF: **Few**$_x$ [**[**$_D$ **e]**$_x$ congressmen]$_y$ $[_{VP}$ **[e]**$_y$ admire Kennedy]

Let us then assume that a version of Quantifier Raising (QR) applies in covert syntax to the quantificational D(eterminer) head, leaving behind a variable as illustrated in (43b). Note that the operations in (43) accomplish in covert syntax what Heim's (1982) NP-Prefixing and Quantifier Construal do. Alternatively, we may also adopt Chomsky's (1995: 202 ff.)) 'Copy plus Deletion' analysis of movement as illustrated in (44).

(44) a. *Copy for Subject Raising*:
LF$_j$: ***[Few congressmen]*** $[_{VP}$ *[Few congressmen]* admire Kennedy]

b. *QR*:
LF$_j$: **Few**$_x$ [**[**$_D$ **e]**$_x$ congressmen] $[_{VP}$ **Few**$_x$ [**[**$_D$ **e]**$_x$ congressmen] admire K.]

c. *LF-deletion under Identity*:
LF$_k$: **Few**$_x$ ~~[[$_D$ e]$_x$ congressmen]~~ $[_{VP}$ ~~Few$_x$~~ **[[**$_D$ **e]**$_x$ **congressmen]** admire K.]

In this analysis, the subject raising copies the internal subject and reconstructs it in the external subject position, as illustrated in (44a). Each quantificational D then undergoes QR, as in (44b). Finally, as illustrated in (44c), part of each operator-variable construction in (44b) is deleted under identity in accordance with the Preference Principle for Reconstruction in (45) below, and derives a single operator-variable construction at LF.

(45) Minimize the restriction in the operator position. (Chomsky (1995: 209))

Under either analysis, QR leaves behind a DP which consists of a variable bound by an operator and an NP to be interpreted as the restrictor of this operator as in (46). (Again, the bound status of the variable is indicated by **a subscript-σ as a mnemonic marker.**)

(46) Few_x .... $[_{DP}\ [_{D}\ e\ ]_{x\sigma}\ [_{NP}$ congressmen]]

In the rest of this work, we will refer to a DP of this type somewhat loosely as '**Bound Trace**' (with the capitals B and T). We then can translate the presence of a quantificational element in (41a–c) into the presence of a 'Bound Trace,' and each such Bound Trace is identified as the antecedent of an E-type pronoun as illustrated in (47a–c).

(47) a. LF: Few_x $[_{DP}\ [_{D}\ e_{x\sigma}\ ]$ **congressmen** $]_y$ admire K. **They**$_y$ are very junior.
b. LF: John owns some_x $[_{DP}\ [_{D}\ e_{x\sigma}\ ]$ **sheep** $]_y$ and Harry vaccinates **them**$_y$ . . .
c. LF: A_x $[_{DP}\ [_{D}\ e_{x\sigma}\ ]$ **dog** $]_y$ came in. **It**$_y$ lay down under the table.

Suppose now that we are indeed correct in hypothesizing that the interpretation of a pronoun involves the covert copying of its antecedent **when no c-commanding relation exists** between them. Recall also that the Bare Output Conditions prohibit us from discriminating, in covert syntax, overt proforms from phonetically-empty proforms involved in ellipsis. We then can identify a point-by-point parallelism between E-type anaphora and the strict identity observed in the ellipsis constructions in (32)–(34). They both involve a DP as part or whole of the antecedent of some anaphoric item to be interpreted elsewhere in the discourse, and this DP consists of: (i) a projection of D (maximal or minimal) as a proform to be interpreted as a bound variable (e.g., $[_{DP}\ his_{X\sigma}]$ in (48a) and $[_{D}\ e_{X\sigma}\ ]$ in (48c) below), and (ii) an NP functioning as the restrictor of the operator (e.g., $[_{NP}$ *teacher*] in (48a) and $[_{NP}$ *congressmen*] in (48c)).[13]

(48) a. VP-Ellipsis in (32): $[_{DP}\ [_{DP}\ \textbf{his}_{x\sigma}\ ]\ [_{NP}$ teacher]]
b. DP-Ellipsis in (34): $[_{DP}\ [_{DP}\ \textbf{zibun}_{x\sigma}\ ]$-no $[_{NP}$ gakusee]]
c. E-type Anaphora in (47): $[_{DP}\ [_{D}\ \textbf{e}_{x\sigma}\ ]\ [_{NP}$ congressmen]]

Since the two phenomena now look parallel, not relating them would strike us as missing some generalization. I therefore would like to propose that our analysis of ellipsis constructions, which we concluded to involve the covert copying of a bound DP-proform, can and should be extended to the analysis of E-type anaphora as well. In particular, I propose that the LF-derivation of an E-type pronoun involves the copying of a Bound Trace showing up earlier within the discourse. The LF-derivation of the example (41a) is illustrated in (49).

(49) a. *QR*:

LF$_i$: Few$_x$ [$_{DP}$ [$_D$ $e_{x\sigma}$] **congressmen**]$_y$ admire K. **They**$_y$ are very junior.

b. *Copy*:

LF$_j$: Few$_x$ *[*$_{DP}$ *[*$_D$ $e_{x\sigma}$ *] congressmen]*$_y$ admire K.

[$_{DP}$ *[*$_D$ $e_{x\sigma}$ *]* ***congressmen*** *]*$_y$ are very junior.

First, as in (49a), the quantificational D undergoes QR in the first clause. Then, as in (49b), the DP-antecedent containing the trace of the raised D, i.e., our 'Bound Trace,' is copied onto the E-type pronoun. The E-type pronoun, thus, comes to be interpreted as a definite description like '**those** congressmen,' in accordance with the Novelty Condition. The copied bound variable of the form *[*$_D$ $e_{x\sigma}$ *]*, in other words, is interpreted on a par with something like a demonstrative determiner used anaphorically, for example, like *that* in English or *so* in Japanese.

We can immediately mention a couple of empirical advantages. First, the contrast as in (50) is known to be recalcitrant to approaches which let E-type pronouns be interpreted based upon pragmatic contexts. Note that both sentences in (50) contain a pragmatically equivalent antecedent clause, and hence are predicted to equally allow the subsequent pronoun to be legitimately interpreted, contrary to the fact.

(50) a. *John has **a wife*** and **she** hates him.
b. **John is married* and **she** hates him. (Evans (1977: 147))

In the syntactic copying approach like ours, the contrast follows naturally, since in (50a), there exists a Bound Trace to be copied as the antecedent of the pronoun after the indefinite undergoes QR, while in (50b), no relevant antecedent DP exists which can be copied onto the pronoun.

The proposed analysis involving DP-copy is also free from the overcopying problem as in (51) associated with other types of syntactic copying approaches like IP-copy by Heim (1990).

(51) A: **A man** jumped off the cliff.
B: **He** didn't jump. **He** was pushed. (Heim (1990: 172))

Heim's IP-Copy would copy and adjoin an antecedent IP as in (52) to *he*, and incorrectly force this pronoun to be interpreted as something like 'a man that jumped off the cliff.'

(52) [$_{IP}$ a man$_1$ [$_{IP}$ t_1 jumped off the cliff]]

The proposed approach also wins a theoretical advantage of maintaining a single syntactic operation of covert copying for the analysis of VP-Ellipsis, DP-Ellipsis, E-type pronouns, and in fact that of 'coreferential' pronouns in general.

4. Minimal variable binding and logical number

In this section, we will see first that the proposed approach to E-type anaphora encounters a potential problem. We will argue that this problem will be solved by postulating a type of economy condition of a general nature. It will then be pointed out that this economy-based solution will permit us to further broaden our empirical coverage thereby providing us with independent motivation for the proposed approach.

4.1 Overgeneration of operators

First, let us observe that (53a) and (53b) are not synonymous.

(53) a. Few congressmen$_x$ admire Kennedy. **They**$_x$ are very junior.
b. Few congressmen$_x$ admire Kennedy. **Few congressmen**$_{y/*x}$ are very junior.

In (53a), the pronoun *they* may be intended to denote the set of individuals defined by *few congressmen* showing up in the previous sentence. The two instances of *few congressmen* in (53b), on the other hand, cannot be intended to define the same set of individuals, although the two distinct sets of individuals they define may end up with overlapping partially or perhaps even completely.

What this contrast implies to the proposed approach to E-type anaphora is that, for the LF-derivation of (53a), we should not allow the covert computational process to duplicate the quantificational antecedent itself, while we still would like to have its Bound Trace to be copied. Since nothing we have postulated so far guarantees such selective application of covert Copy, we have a problem of overgenerating an operator at LF, as illustrated in (54).

(54) a. *Copy*:
LF$_j$: Few congressmen admire Kennedy. **Few congressmen** are very junior
b. <u>*QR*</u>:
LF$_K$: **Few**$_x$ [$_{DP}$ [$_D$ e]$_{x\sigma}$ congressmen] admire Kennedy, and
Few$_x$ [$_{DP}$ [$_D$ e]$_{x\sigma}$ congressmen] are very junior

In this derivation, a covert operation first copies the quantificational DP *few congressmen* as in (54a) and then QR applies to both of the original and the duplicate of this operator as in (54b), yielding an LF-representation identical to that for (53b) – an undesirable consequence.

The restriction observed in (53b) actually seems to be a quite general crosslinguistic restriction imposed on the identity of two **base-generated** morphologically identical operators, as illustrated by the examples in (55).

(55) a. *subete-no syoosya-ga* **aru daigaku-no subete-no gakusee**$_x$-*o* *husaiyoo-ni-sita*
all trading firm-NOM one university-GEN all **student**-ACC didn't.hire
ga, ikutukano ginkoo-wa **aru daigaku-no subete-no gakusee**$_{y/*x}$-*o* *saiyoosita.*
but, some banks-TOP one university-GEN all student-ACC hired
'While all trading firms declined to hire any of the students of one university, some banks hired all of the students of **some other university.**'
b. **Many people**$_x$ were unhappy, and **many people**$_{y/*x}$ left.
c. **Someone**$_x$ called. **Someone**$_{y/*x}$ didn't leave a message.
d. He has **a cat**$_x$ and she hates **a cat**$_{y/*x}$. (cf. The Novelty Condition)

In the Japanese example (55a), for instance, the two instances of *aru daigaku-no subete-no gakusee* 'all the students of one university' must be intended to denote two different sets of students from distinct universities.

The examples in (55) again contrast with those in (56) below, in which a proform succeeds in establishing an anaphoric relation with a quantificational antecedent.

(56) a. *subete-no syoosya-ga* **aru daigaku-no subete-no gakusee**$_x$-*o*
husaiyoo-ni-sita ga, ikutukano ginkoo-wa **[e$_x$]** *saiyoosita.*
'While all trading firms declined to hire any of the students of one university, some banks hired **them.**'
b. **Many people**$_x$ were unhappy, and **they**$_x$ left.
c. **Someone**$_x$ called. **He**$_x$ didn't leave a message.
d. He has **a cat**$_x$ and she hates **it**$_x$.

It therefore seems to be generally true that more than one instance of a morphologically identical operator cannot be intended to denote an identical set in a single discourse. Here, I would like to propose that this generalization follows from the economy constraint imposed on operators as in (57).

(57) *Minimal Variable Binding*:
Variable binding is minimal – a 'single' operator establishes an operator-variable binding **only once in a derivation.**

The notion 'single' operator referred to in (57) is defined as in (58).

(58) *'Single' Operator*:
One or more instance of a morphologically identical operator constitutes a 'single' operator if they are intended to define an identical set.

When we assume that each instance of a quantificational element must undergo QR and binds its variable at LF, this economy constraint will yield the effect of prohibiting more than one instance of a 'single' operator from showing up at LF, whether they are base-generated or derived by the application of covert Copy. We may then consider that the covert copying of the quantificational DP in (54a) is in fact legitimate, but the constraint (57) rules out the multiple operator-variable relations established by a 'single' operator as in (54b). We thus can free ourselves from the problem of overgenerating operators at LF in the analysis of E-type anaphora.[14]

The Minimal Variable Binding in (57) can be independently motivated in an interesting way. Observe, first, the paradigm in (59) below. (59a) indicates that a universally quantified element *every boy* must be treated as singular. (59b), on the other hand, suggests that *every boy* in fact is plural. Making the situation even more complicated, (59c) illustrates that *every boy* can be ambiguous between singular and plural. A universally quantified element, in other words, exhibits rather unpredictable restriction as well as flexibility in its number agreement with other items.[15]

(59) a. **Every boy** { **is** / ***are** } happy.
  b. **Every boy** left. { ***He** / **They** } must be angry.
  c. **Every boy** knows that { **he** / **they** } should apologize.

A careful examination of this paradigm, however, will provide us with the following observations and generalizations. First, this paradigm in fact involves two distinct types of number agreement – 'inflectional' agreement in (59a) and 'referential' agreement in (59b) and (59c). It may be said, in other words, that *every boy* exhibits flexibility in referential agreement but not in inflectional agreement. This point can be confirmed when we observe that (59c) becomes ungrammatical when we alter the inflectional agreement while leaving the rest of the sentence intact as in (60).

(60) *Every boy **know** that { he / they } should apologize.

The contrast between (59b) and (59c) further suggests that the flexibility in referential agreement is permitted **only when the antecedent c-commands the pronoun.**[16] Furthermore, when we replace the quantificational subject *every boy* in (59c) with a non-quantificational subject as in (61a) and (61b) below, the referential agreement also becomes static. This suggests that the quantificational force of the antecedent is crucial in permitting the flexibility of referential agreement.

(61) a. **That boy** knows that { **he** / ***they** } should apologize.
  b. **Those boys** know that { **they** / ***he** } should apologize.

All these observations follow when we extend the notion of linguistic number in the way described below, and combine it with the Minimal Variable Binding in (57).

First, we recognize what we will refer to as 'overt number,' which is the number associated with a 'nominal' lexical form in the lexicon (or in the numeration, if one opts for such an approach). This is perhaps the standard notion of linguistic number for syntacticians, and it has traditionally been regarded as relevant to both inflectional and referential agreement.

In addition to overt number, we recognize what we will refer to as 'logical number.' Logical number is activated by operator-variable binding and realized on the DP containing the trace of a raised D as a variable, and hence only quantificational elements can exhibit logical number in addition to its overt number. Since it is activated for the first time at LF, it is relevant only to meanings, contrary to overt number, which may be relevant to both meanings and forms.[17] Logical number therefore plays a role in referential agreement but not in inflectional agreement, which has PF properties as its indispensable aspect. Overt number, on the other hand, is available both at PF and LF, and it can play a role in both inflectional and referential agreement.

If a nominal element is non-quantificational, overt number generally is the only possible type of number it may have, and it never exhibits flexibility in either inflectional or referential number agreement, as observed in (61).[18] A lexical form of a quantificational nominal also has its overt number. *Every boy*, for example, is associated with singular number just like the non-quantificational *that boy* is. In the proposed number system, however, this will not be the end of the story. While a quantificational element exhibits overt number before it undergoes QR, it exhibits, or more precisely its Bound Trace exhibits, logical number after it undergoes QR. As a result, if there exists discrepancy between the overt number and the logical number of a quantificational element, a possibility arises for some flexibility in its referential agreement.

I believe that the flexible referential agreement observed in (59c) above is one such case, in which either of the **singular overt** number associated with the lexical form *every boy* and the **plural logical** number associated with its Bound Trace may play a role in the referential agreement involving this quantificational antecedent. How does this discrepancy between the overt number and logical number of *every boy* arise? In particular, what is the source of its plural logical number? The answer to this question seems to lie in the function of the operator-variable binding established by the application of QR as in (62).

(62) LF: **Every**$_{x}$ [$_{\mathbf{DP}}$ [$_{\mathbf{D}}$ **e**]$_{x\sigma}$ **boy**] left.
↑__ QR __|

Roughly speaking, this operator-variable binding establishes plural eventualities by letting the quantifier *every* pick out all the members of a presupposed set defined by its restrictor *boy* in a given pragmatic context, which, at least in default cases, is **non-empty and non-singleton**. Subject to slight modification below, we ascribe the non-singleton status of such a presupposed set as the source of the plural logical number of a universally-quantified nominal expression like *every boy*.

When we combine this extended notion of number with the Minimal Variable Binding (57) adopted above, we can capture the otherwise puzzling agreement facts in the paradigm (59) quite straightforwardly. First, the static nature of the singularity of *every boy* observed in (59a) can be ascribed to its overt number, which is the only number relevant to the PF-reflection of inflectional agreement. Second, we can also capture the flexibility of referential agreement in (59c), in which *every boy* c-commands the pronouns. As illustrated in (63a) below, the singular pronoun *he* may agree with the lexical form *every* *boy*, which exhibits singular overt number. While the plural pronoun *they* has no chance to legitimately agree with *every boy* itself, after this quantificational antecedent undergoes QR as in (63b), this pronoun may agree with its Bound Trace, which exhibits plural logical number. (Dotted lines indicate referential agreement.)

(63) a. *Referential agreement with* **overt** *number* (**before QR**):
LF: **Every boy** knows that **he** should apologize.

b. *Referential agreement with* **logical** *number* (**after QR**):
LF: [Every$_x$ [**[$_{DP}$ [$_D$ e]$_{x\sigma}$ boy]** knows that **they** should apologize]]
QR

Finally, the absence of similar flexibility in referential agreement in (59b) also follows naturally. Recall that the pronouns are not c-commanded by their quantificational antecedent *every boy* in (59b), which is the structural condition for the application of covert Copy, as we have seen before. First of all, when the pronoun is plural as in (64a) below, legitimate referential agreement can take place only when the copied antecedent is plural. This situation can arise when *every boy* undergoes QR as in (64b), and the Bound Trace, which is plural, is identified with the plural pronoun *they* as in (64b), and copied as in (64c).

(64) a. **Everybody** left. **They** must be angry.
b. *QR*:
LF$_i$: **Every$_x$ [$_{DP}$ [$_D$ e]$_{x\sigma}$ boy]** left. **They** must be angry.
QR

c. *Copy*:
LF$_j$: Every$_x$ *[$_{DP}$ [$_D$ e]$_{x\sigma}$ boy]* left. ***[$_{DP}$ [$_D$ e]$_{x\sigma}$ boy]*** must be angry.

Crucially, when the Bound Trace is copied as in (64c), the Minimal Variable Binding (57) is satisfied, and the derivation converges.

When the involved pronoun is singular, as in (65a) below, on the other hand, it must agree with the lexical form *every boy*, which is singular, rather than with its Bound Trace, which is plural. The copying of the operator in its lexical form therefore is required, as in (65b).

(65) a. LF$_i$: **Every boy** left. **He** must be angry.

b. *Copy*:
LF$_j$: *Every boy$_x$* left. ***Every boy$_x$*** must be angry.

c. *QR*:
LF$_k$: **Every$_x$** [$_{DP}$ [$_D$ e]$_{x\sigma}$ boy] left. **Every$_x$** [$_{DP}$ [$_D$ e]$_{x\sigma}$ boy] must be angry.

This derivation will eventually crash, however, failing to satisfy the Minimal Variable Binding (57) when both the original and the duplicate of *every body* undergo QR and establish more than one instance of operator-variable binding by a 'single' operator, as in (65c).

One of the anonymous reviewers of this volume poses an interesting and relevant question as follows – if the ellipsis constructions and the examples with the E-type anaphora are indeed parallel, as pointed out above, why is it that the non-elliptical sentence involving a base-generated pronoun *his* in the second clause of (66) below (= (21) above) does not permit a strict variable while its elliptical counterpart (67) (= (16)) does?

(66) In our school, **every student**$_x$ [$_{VP}$ respects his$_x$ teacher], but unfortunately, the **PRÍNcipal DÓESn't** [$_{VP}$ ↓*respect *his$_x$ teacher*↓]. (= (21))

(67) In this school, **every student**$_x$ [$_{VP}$ respects **his**$_x$ teacher], but unfortunately, the principal doesn't [$_{VP}$ e]. (= (16))

The approach we have just argued for allows us to give a straightforward answer to this question. As has already been pointed out above, it is impossible in (66) for the base-generated pronoun *his* in the second clause to be directly bound by the quantified element *every student*, with no c-command relation holding between them. If covert reconstruction is to apply to associate them, it must copy the singular *every student* rather than the plural Bound Copy onto *his*, which would fail to satisfy the Minimal Variable Binding just as in (65). We then predict that, if the pronoun to be base-generated in the second clause of (66) is the plural *their*, the sentence permits a strict variable and becomes grammatical, involving LF-copy of a Bound Trace just as in (64). The example in (68) below demonstrates that such indeed is the case. (The second clause in (68) requires prosodic reduction of the VP (enclosed by ↓___↓) following emphatic stress of the focused items (indicated by capitalized syllables), presumably to justify the repetition of redundant information carried by the VP in the second clause (Kitagawa (1999a)). Note that similar prosody does not improve (66).)

(68) In our school, **every student**$_x$ [$_{VP}$ respects his$_x$ teacher], but unfortunately, the **PRÍNcipal DÓESn't** [$_{VP}$ ↓*respect **their**$_x$ teachers*↓].

Thus, the proposed covert copying approach incorporating the Minimal Variable Binding (57) and the extended number system permits us to capture the otherwise mysterious and arbitrary flexibility of number agreement observed in the paradigm (59) (as well as the contrast between (66) and (67)), thereby providing an independent piece of motivation for the notion 'Minimal Variable Binding.'

4.2 Indefinites vs. ∀ and Neg

The extended number system we have adopted has further virtues. It is often pointed out in the literature that indefinites need not observe a 'scope-island,' while other quantificational elements like *every* and *no* must, as illustrated by the paradigm in (69).[19]

(69) a. **A** dog$_x$ came in. **It**$_x$ lay down under the table.
b. **Every** dog$_x$ came in. ***It**$_x$ lay down under the table.
c. **No** dog$_x$ came in. ***It**$_x$ lay down under the table. (Heim (1982: 13))

As has been noted sporadically in the literature,[20] and as has been observed in (59b) above, however, *every* and *no* also need not observe a 'scope-island' when the pronoun is plural. Some relevant examples are shown in (70).

(70) a. **Every** dog$_x$ came in. **They**$_x$ lay down under the table.
b. [A detective trying to prove that Mary is lying says]:
The truth is that John wrote **no** article$_x$.
Mary therefore cannot possibly have read **them**$_x$.
c. If John owns **no** sheep$_x$, there is no way for Harry to vaccinate **them**$_x$.

Some researchers disregard such examples, claiming that plural pronouns involve some anaphoric relation distinct from E-type anaphora. Chierchia (1995: 4), for instance, states that '... every quantified noun phrase can make salient a set of entities (roughly, the set associated with its head noun). This set can then be referred to in subsequent discourse.'

Simply stating that the set associated with the head noun can be referred to by a plural pronoun, however, leaves many important questions unanswered. For instance, the second clause in (71) expresses multiple eventualities quite naturally as its primary reading.

(71) Every dog$_x$ came in one by one, and **they**$_x$ lay down where they$_x$ were supposed to.

Therefore, when we interpret (71), it is quite natural for us to imagine a chain of events in each of which a dog comes in and lies down wherever it wanted to, each dog ending up in a different place. When a plural nominal head is overtly expressed in the second clause as in (72) below, on the other hand, it is noticeably less natural to make a primary reading out of a similar distributive interpretation.

(72) Every dog$_x$ came in one by one, and **the dogs**$_x$ lay down where they$_x$ were supposed to.

The distributive interpretation in the second clause of (71) can be naturally ascribed to the existence of quantification in the first clause if the Bound Trace of *every / no* is copied onto *they.* If each plural pronoun in (70) simply refers to a set associated with the head nominal of its antecedent, on the other hand, the contrast between (71) and (72) would remain mysterious. This approach would also leave it unexplained why the head nouns do not yield similar plurality when they are quantified by existential quantifiers, as in (73).[21]

(73) a. **A** dog$_x$ came in. ***They**$_x$ lay down under the table.
b. **Some** dog$_x$ came in. ***They**$_x$ lay down under the table.

We can, on the other hand, capture all the facts in (69)–(73) in terms of referential number agreement when we postulate the logical number as in (74) below for the Bound Traces of the quantificational elements involved in these examples.

(74)

	Overt Number	Logical Number
Every / No	Singular	**Plural**
Some / A	Singular	**Singular**

Though in the opposite way, a negative D *no* behaves exactly like *every*, and picks out **none** of the members of the presupposed non-empty, **non-singleton** set. It therefore exhibits plural logical number just like *every*. The existential D's *some* and *a*, on the other hand, do not involve any such non-singleton set in a given pragmatic context. They therefore exhibit singularity in both their overt and logical numbers. In all of (69)–(73) above, therefore, DPs containing *every* or *no* must referentially agree with a plural pronoun, while those containing *a* or *some* must referentially agree with a singular pronoun. A crucial basis of our analysis again is the corollary of the Minimal Variable Binding we have discussed–that is, these examples necessarily involve the copying of Bound Traces rather than the quantificational antecedents. Thus, the otherwise puzzling contrast between *every/no* and *some/a* follows straightforwardly in the proposed approach incorporating the Minimal Variable Binding and the extended number system.

Earlier, we identified, as the source of the plural logical number, a non-singleton status of a set presupposed by a quantifier. The following example suggests that we need to be a little more precise in making such a statement:

(75) Pick up **every book** on the desk, *if there's any*, and bring { **them** / ***it** } back to me.

Note that the expression *if there's any* indicates that the speaker does not presuppose the **actual** existence of any set of books on the desk. Yet, the plural pronoun *them* can be still anaphoric to *every book*. Incorporating this observation, we revise our generalization as in (76).

(76) The logical number of an operator is plural if the '**candidate**' set it presupposes in a given pragmatic context is non-empty and non-singleton.

The intuitive idea behind this generalization is that the use of *every* and *no* presupposes the existence of a 'candidate' or 'potential' set of plural entities out of which the quantifiers select a designated quantity of the members, even if it does not presuppose the actual existence of such entities.[22]

Finally, there are cases as in (77) below, in which E-type pronouns may show up **either** as singular **or** plural.

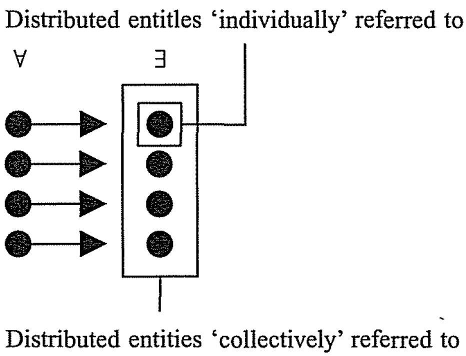
(77) *Every* student turned in **a paper**$_x$.
a. **It**$_x$ went into his or her folder.
b. **They**$_x$ were all identical. (Heim (1990))

The phenomenon here strikes us as contradictory to the number agreement analysis provided above, in which we assumed that the Bound Trace of an existential quantifier is singular in its logical number. If we examine the example in (77) carefully, however, we notice that there is an extra factor involved. The first clause contains a logically plural quantificational expression *every student* c-commanding the indefinite as the antecedent of the E-type pronouns. On the contrary, when we replace *every student* in (77) with the logically singular *some student* as in (78) below, a plural pronoun can no longer be anaphoric to the indefinite.

(78) **Some** student turned in a paper$_x$. { **It**$_x$ / ***They**$_x$ } went into her folder.

The source of plurality in (77b), in other words, indeed seems to be the presence of a higher logically plural quantificational expression. How can this extra factor turn the singular Bound Trace of an existential quantifier into plural? Although I am not ready to offer any definite answer to this question, it seems plausible for us to tentatively ascribe this phenomenon to the distributivity involved in the scopal interaction between universal and existential quantifiers. That is, when an existential quantifier takes its own scope within the scope of a universal quantifier, the logically singular Bound Trace of an existential quantifier may distribute under the logically plural universal quantifier, and such distributed entities may be referred to not only 'individually' as singular but also 'collectively' as plural, as graphically illustrated in (79).

(79)



We still maintain, in other words, our position that the Bound Trace of an existential quantifier per se is logically singular.

We can confirm that this insight is leading us in the right direction when we examine the scope interaction of two quantificational expressions in an example like (80).

(80) Every girl falls in love with **some prince**.
  a. **He**$_x$ is distinctively noble and breath-takingly handsome.
  b. **They**$_x$ are distinctively noble and breath-takingly handsome.

Just as in the case of (77) above, the existentially quantified element in (80) can induce either singular or plural E-type anaphora when it is interpreted distributively under the scope of the universal quantifier. On the contrary, if we eliminate such distributivity by letting the existential quantifier in (80) take scope higher than that of the universal quantifier, perhaps with an emphatic stress on *some*, plural E-type anaphora as in (80b) no longer seems to be permitted. It therefore seems possible to ascribe the plurality an indefinite exhibits as in (77) while maintaining its logical singularity as postulated in (74).

5. From E-type anaphora to donkey anaphora

When we adopt a traditional view and regard the indefinite determiner *a*(*n*) as an existential quantifier, donkey sentences share all of the properties of E-type pronouns stated above in (42). That is, they have **quantified** DPs as their **antecedents**, they are **not c-commanded** by their antecedents, yet they are successfully interpreted as **definite** descriptions. It seems only natural therefore to extend the proposed analysis of E-type anaphora to donkey anaphora, making a precedent of researchers like Cooper (1979), Lappin (1988/89) and Heim (1990). We thus propose that donkey anaphora involves covert copying of the 'Bound Trace' of an existential quantifier, as illustrated by the derivation in (81).

(81) a. *QR*:

LF$_i$: [Every farmer who **a**$_X$ owns [$_{DP}$ **[**$_D$ **e**$_{x\sigma}$ **]** donkey]$_y$] beats it$_y$

b. *Copy*:

LF$_j$: [Every farmer who a$_X$ owns [$_{DP}$ [$_D$ e$_{x\sigma}$] donkey]$_y$] beats **[**$_{DP}$ **[**$_D$ **e**$_{x\sigma}$ **] donkey]**$_y$

As in the case of E-type anaphora, we can immediately account for the contrast as in (82) below in this approach, which remains recalcitrant if the main properties of donkey anaphora are to be reduced to pragmatic factors.

(82) a. [*Every man who has* ***a wife***] sits next to **her**.
b. *[*Every married man*] sits next to **her**. (Heim (1990: 165))

In (82a), there exists a Bound Trace to be copied as the antecedent of the pronoun after the indefinite determiner undergoes QR, while in (82b), no relevant antecedent exists which can be copied onto the pronoun. In what remains, I will first provide a brief overview of donkey anaphora, and then critically examine some alternative approaches, comparing them to our proposal when relevant. Such a process hopefully will help us clarify which of the observations concerning donkey anaphora should or should not be captured by formal grammatical devices.

In addition to the characteristics mentioned above, donkey anaphora is said to exhibit a few more puzzling properties. First, 'maximality' of interpretation is said to arise – an indefinite expression showing up in a donkey sentence as in (83) is interpreted as universally quantified as indicated by the translation there (henceforth ∀-readings) (Geach (1962)).

(83) [Every farmer who has **a donkey**] beats **it**.
'Every donkey-owning farmer beats **every donkey** that s/he owns.'

The source of the universal force of the pronoun *it*, however, is not at all evident since its antecedent *a donkey* has traditionally been regarded as possessing existential force. Second, an 'accessibility' constraint is recognized – while a donkey pronoun need not be c-commanded by its indefinite antecedent, it still must be c-commanded by the lowest quantifier that c-commands that indefinite (Haïk (1984)):

(84) *[[*Everyone* who owns **a donkey**] came], and Mary bought **it**.

In (84), contrary to (83), the donkey pronoun *it* fails to be c-commanded by the universal quantifier *everyone*(, which c-commands the indefinite expression *a donkey*,) and as a result, it fails to function as a donkey pronoun.

One of the most celebrated analyses of donkey anaphora involves (syntactic) 'unselective binding' as in (85) (Kamp (1981), Heim (1982)). See also May (1985).

(85) $\forall_{\mathbf{x},\,\mathbf{y}}$ [farmer (x) $\wedge$ donkey (**y**) $\wedge$ own (x, **y**)] [beat (x, **y**)]
'For **every pair** < **x, y** > such that x is a farmer, y a donkey, and x owns y, x beats y'

As indicated by the translation in (85), this analysis derives the universal quantification over a pair of variables, which solves the scope problem and yields maximality to the interpretation of indefinites. Despite the popularity of the syntactic mechanism involved in this analysis, however, its problems in the analysis of donkey anaphora is also well-known. For example, Pelletier and Schubert (1989) have pointed out that an existential rather than universal interpretation of the indefinite is possible in a donkey sentence like (86).

(86) Every person who has **a dime** will put **it** in the meter.

A quite natural interpretation of this sentence is 'Every person who has a dime will put in the meter **one of his/her dimes**', which clearly permits the existential readings (henceforth ∃-readings) of the donkey pronoun, and hence does not involve maximality. Maximality, in other words, is not the norm but only a possibility associated with donkey anaphora. This state of affairs in (86), however, is totally unexpected by unselective binding of *a dime* and *it* by the universal quantifier *every*.

Another problem the unselective binding analysis encounters is what is known as a 'proportion problem' (Heim (1990)). In a donkey sentence as in (87a), for example, the operator *most* unselectively quantifying over a pair of variables would yield the interpretation in which donkey beating takes place with respect to **most** farmer-donkey **pairs**, as represented in (87b). (87a) therefore would be incorrectly true if nine farmers that own one donkey each don't beat it and one farmer that owns fifty donkeys beats them all.

(87) a. Most farmers that have a donkey beat it.
b. **most** $_{x, y}$ [farmer (x) $\wedge$ donkey (y) $\wedge$ x own y] [x beats y]

Because of these and other problems the unselective binding encounters, I will refrain from regarding it as a viable approach to donkey anaphora.

Let us now direct our attention to the 'dynamic binding' approach, in particular, that proposed by Chierchia (1995). In a nutshell, this approach advocates a semantic process of bringing a sentence in a discourse into a previous sentence, as illustrated in (88).

(88) a. [$S_1' \wedge p$]
b. [$S_1' \wedge p$] $\wedge$ *[$S_2' \wedge p$]* $\Rightarrow$ [$S_1' \wedge S_2' \wedge$ **p**]

Crucially, discourse sequencing is regarded here as an operation of conjunction, which has the effect of replacing a propositional variable *p* with a subsequent discourse. Chierchia claims then that the ∃-readings of donkey pronouns are derived semantically when the conservativity of determiners as illustrated in (89) below interacts with dynamic binding and induce 'dynamic conservativity.'

(89) D (P)(Q) $\leftrightarrow$ D (P)(P $\wedge$ Q)
e.g.) { Every / Some / No } man smokes iff { every / some / no } man is a man who smokes.

In the tripartite representation of a donkey sentence as in (90a) below, for example, the general conservativity of determiners as in (89) allows the indefinite *a dime* be reconstructed **within the nuclear scope**, as illustrated in (90b). The dynamic property of a conjunction operator $\wedge$ then brings the nuclear scope *puts it in the meter* into the reconstructed restrictor, as illustrated by the arrow there, and permits the donkey pronoun to be bound by existential quantifier in its domain.

(90) Every person who has **a dime** puts it in the meter.
a. Every (person who has a dime) (puts it in the meter)
b. Every (person who has a dime) ([person who has
a dime $\wedge$ *p*] $\wedge$ ***[puts it in the meter]***)

As a result, the representation will yield the interpretation 'Every person who has a dime is person who has a dime and puts it in the meter,' in which the indefinite as an existential quantifier binds the donkey pronoun outside the domain of the universal quantifier and yields the ∃-reading. The dynamic binding derivation of ∃-readings can be characterized by the following properties. First, since this approach treats the indefinite as existentially quantified expression and it shows up outside the domain of the universal quantifier, there is no room for maximality presuppositions to arise. Second, ∃-readings of donkey anaphora arise when the nuclear scope containing the donkey pronoun is subordinated into the reconstructed restrictor, which originally was the domain of the universal quantifier. Donkey pronouns therefore must always be contained in the nuclear scope and necessarily satisfy the accessibility requirement when they yield ∃-readings. Finally, the dynamic binding derivation of ∃-readings of donkey anaphora is due to purely semantic operations, and therefore is insensitive to pragmatic factors.

Having acknowledged the possibility of anaphora across inaccessible domains as in (91) below, on the other hand, Chierchia pronounced that the dynamic binding approach must also be supplemented by what he labels as the 'E-type strategy,' in which E-type pronouns are regarded as **functions** from individuals to individuals, where the nature of function is **contextually** specified.

(91) a. Either [Morrill Hall doesn't have **a bathroom**] or **it** is in a funny place. (Barbara Partee)
b. It is not true that [John doesn't have **a car**]. **It** is parked outside.
d. [John does**n't** have **a car** any more]. He sold **it** last month.

(92a–c) illustrate how the function of each E-type pronoun in (91) can be determined.

(92) a. Either [Morrill Hall doesn't have a bathroom] or **f (Morrill Hall)** is in a funny place.
– **f**: a function from places into bathrooms located in those places
b. It is not true that John doesn't have a car. **f (John)** is parked outside.
– **f**: a function from people into their cars
c. John doesn't have a car any more. He sold **f (John)** last month.
– **f**: a function from people into the car they used to have

Chierchia then claims that ∀-readings of donkey anaphora are made possible by this pragmatic E-type strategy. In a donkey sentence like (93a) below, for example, the pronoun as function is **number neutral** and can have a **plural** entity, as indicated in (93b).

(93) a. [Every man who has **a donkey**] beats **it**.
b. $\forall x\ [[\ \mathbf{man\ (x)}\ \wedge\ \exists y\ [\ \text{donkey}\ (y)\ \wedge\ \text{has}\ (x, y)\ ]]\ \rightarrow\ \text{beats}\ (x, \mathbf{f\ (x)})]$
– **f**: a function from men into the **donkey** (or **donkeys**) they own
'Every man beats **the donkey or donkeys** he owns.'

As a result, the donkey pronoun can take the **maximal** set of donkeys owned by each man as its value, and yield the ∀-readings. The E-type strategy, in other words, is claimed to be responsible for **maximality** presuppositions in donkey anaphora contexts. Since the value of donkey pronouns is determined pragmatically, the E-type strategy is also predicted **not** to have to satisfy accessibility constraints.

The correlations between the type of interpretation and the compliance/rejection of accessibility constraints predicted in the dynamic binding approach, however, do not seem to always obtain. In particular, the ∃-readings of donkey anaphora seems to be possible even when the accessibility condition is not satisfied:

(94) a. [Every person who had **a dime**] asked the secretary, and she put **it** in the meter.
b. John interviewed [every professor in our department who submitted **a paper** to LI], and wrote an article about their experiencing **its** rejection at least once.

Note that the sentence in (94a) is felicitous even when the person handed to the secretary 'one of the dimes he or she had.' Likewise, the professors mentioned in (94b) certainly may have some of their papers accepted. Such a breakdown of the correlation between the availability of ∃-readings and the satisfaction of the accessibility constraint casts doubt on the derivation of ∃-readings based upon the conservativity of determiners in the dynamic binding approach.

Attempting to support his E-type strategy and at the same time to argue against a syntactic copying approach, Chierchia also discusses the possibility of a single donkey pronoun's being anaphoric to two NPs in a coordinate construction as in (95). (We suppress the DP-analysis momentarily here in order to synchronize our discussion with that of Chierchia's.)

(95) a. [$[_{NP}$ Every boy that has **a dog**] and $[_{NP}$ every girl that has **a cat**]] will beat **it**. (Chierchia (1995: 116))
b. $[\ \forall x\ [\ \mathbf{boy\ (x)} \wedge \exists y\ [\ \text{dog}\ (y) \wedge \text{has}\ (x, y)] \rightarrow \text{beat}\ (x, \mathbf{f(x)})] \wedge$
$[\ \forall x\ [\ \mathbf{girl\ (x)} \wedge \exists y\ [\ \text{cat}\ (y) \wedge \text{has}\ (x, y)] \rightarrow \text{beat}\ (x, \mathbf{f(x)})]$

He claims that the E-type strategy can handle this mismatch since, as illustrated in (95b), the pronoun *it* can be interpreted as the union of the two **number neutral** functions – one function from boys into dogs, and another function from girls into cats. The copying approach, on the other hand, allegedly has a hard time handling this anaphoric relation because two distinct indefinites would have to be copied onto a singular pronoun.

I believe, however, that the conclusion here has been drawn prematurely since the sentence in (95a) in fact is subject to two distinct syntactic analyses, and each construction seems to have distinct tolerance for the mismatch in question. As illustrated in (96a) below, the sentence can be analyzes as involving NP-coordination. At the same time, the same string of words can be analyzes as involving IP-coordination with its I' elided as in (96b).

(96) a. *NP-coordination*:
$[_{NP}$ Every boy that has **a dog**] and $[_{NP}$ every girl that has **a cat**] // $[_{I'}$ will beat **it**].
b. *IP-coordination*:
$[_{IP}$ $[_{NP}$ Every boy that has **a dog**] $[_{I'}$ **e**]] // and $[_{IP}$ $[_{NP}$ every girl that has **a cat**] $[_{I'}$ will beat **it**]].

Some speakers even report that each of the two constructions is associated with a distinct prosodic pattern, each involving a distinct position for a major pause as indicated by // in

(96a–b), respectively. Intuitively, the donkey anaphora in question seems to be permitted only when the sentence is 'distiributively' interpreted involving more than one eventuality, which is also reflected by Chierchia's semantic representation in (95b). Note then, that in the IP-coordination in (96b), the elided I' (which contains a donkey pronoun) can be reconstructed at LF as in (97).

(97) LF: [$_{IP}$ [$_{NP}$ Every boy that has a dog] **[$_{I'}$ will beat it]**] and [$_{IP}$ [$_{NP}$ every girl that has a cat] [$_{I'}$ will beat it]]

In this LF-representation, donkey anaphora is permitted in **each conjunct**, and the problem of mismatch disappears. The LF representation here in fact straightforwardly reflects our intuition that distributivity induced by plural eventuality must be involved for the donkey pronoun in (95a) to be successfully interpreted.

We can demonstrate that the obviation of mismatch in question and the availability of IP-coordination correlate with each other. With the addition of two distinct (and incompatible) sentential adverbs as in (98a) below, we can force the sentence in (95a) to have IP-coordination involving I'-Ellipsis, which is interpreted on a par with the non-elliptical sentence (98b).[23]

(98) a. [$_{IP}$ ***Definitely,*** [$_{NP}$ every boy that has a dog] **[$_{I'}$ e]**] // and
[$_{IP}$ ***probably,*** [$_{NP}$ every girl that has a cat]] [$_{I'}$ will beat it]].
b. [$_{IP}$ ***Definitely,*** [$_{NP}$ every boy that has a dog] **[$_{I'}$ will beat it]**] // and
[$_{IP}$ ***probably,*** [$_{NP}$ every girl that has a cat] [$_{I'}$ will beat it]]

In the IP-coordination in (98a), the mismatch between the donkey pronoun *it* and two indefinite NPs (*a dog* and *a cat*) can be clearly obviated.

We can, on the other hand, ensure the involvement of NP-coordination with plural agreement between the coordinated subject and the verb as in (99).

(99) ?*[$_{NP}$ [Every boy that has a dog] and [every girl that has a cat]] // [$_{I'}$ **are** beating **it**].

In this sentence, the distributive interpretation associated with plural eventuality is not available, and the obviation of mismatch is not permitted. This makes a sharp contrast with the sentence involving singular agreement (and hence IP-coordination) as in (100) below.

(100) [$_{IP}$ [$_{NP}$ Every boy that has a dog] */$_{I'}$ e /*] // and [$_{IP}$ [$_{NP}$ every girl that has a cat] [$_{I'}$ { beats / **is** beating } **it**]].

We can, in other words, also demonstrate that the obviation of mismatch in question and NP-coordination must be dissociated.

The same points can be even more clearly demonstrated when we examine Japanese, in which IP- (or possibly VP-)coordination and NP-coordination can be overtly distinguished, and they contrast in regard to donkey anaphora. In (101) below, in which

IP-coordination with I'-Ellipsis is signaled by the presence of two nominative markers *-ga*, donkey anaphora is quite readily accepted.

(101) *IP-coordination*:
Tuusan-syoo-kara-no-otassi-de, [$_{IP}$ [$_{NP}$ **arubaito-no gakusee**-*ni*
MITI-from-GEN-notification-with, working student-DAT
denwa-ban-o saseteita subeteno zimusyo]-**ga** *[*$_{I'}$ *e]*], *sosite*
made.receive.phone.calls all offices]-NOM and
[$_{IP}$ [$_{NP}$ **huriitaa**-*ni miseban-o saseteita subeteno konbini*]-**ga**
part-timer-DAT **made**.work.as.salesclerk all convenient.store-NOM
[$_{I'}$ **soitu**-o *kubinisinakya-naranaku-natta rasii*]].
that.brat-ACC had.to.fire I.heard
'Because of the notification from the Minister of International Trade and Industry (MITI), all the offices that had **a working student** receive phone calls, as well as all the convenient stores that had **a part-timer** as a salesclerk had to fire **him**, I heard.'

In (102) below, in which NP-coordination is signaled by the presence of the conjunct *to* and a single nominative marker *-ga*, on the other hand, donkey anaphora seems much harder to obtain. As a result, the pronominal *soitu* 'the/that brat' must be pluralized to make the sentence natural.

(102) *NP-coordination*:
Tuusan-syoo-*kara-no-otassi-de,* [$_{NP}$ [$_{NP}$ **arubaito-no gakusee**-*ni*
MITI-from-GEN-notification-with, working student-DAT
denwa-ban-o saseteita subeteno zimusyo]-**to**
made.receive.phone.calls all offices-and
[$_{NP}$ **huriitaa**-*ni miseban-o saseteita subeteno konbini*]]-**ga**
part-timer-DAT **made**.work.as.salesclerk all convenient.store]-NOM
{ ***soitu**-*o* / ^{ok}**soitu***ra-o* } *kubinisinakya-naranaku-natta rasii*
that.brat-ACC / those.brats-ACC had.to.fire I.heard

I am therefore inclined to conclude that the donkey anaphora in (95a) is made possible by the covert reconstruction of a donkey pronoun involved in I'-Ellipsis, and that there is no need to make an appeal to the pragmatic E-type strategy to permit this construction.

Finally, let us observe the sentence in (103) below, in which donkey anaphora is permitted, again, when the involvement of two eventualities is detected.

(103) [[Every man who had **a dime**] and [every woman who had **a quarter**]] put **it** in the meter.

What attracts our attention most here is that this sentence can be interpreted with a rather clear ∃-reading 'Every man who had a dime put **one of his dimes** in the meter, and every woman who had a quarter put **one of her quarters** in the meter.' According to Chierchia, the success of donkey anaphora in this coordinated construction requires the adoption of the pragmatic E-type strategy. The possibility of ∃-readings, on the other hand, would call for the purely semantic dynamic binding analysis. This contradiction suggests that there is something wrong with adopting either one or both of the dynamic binding and the pragmatic E-type strategy. We thus have reasonable doubt on Chierchia's hybrid approach to donkey anaphora.

Earlier, we proposed that all of VP-Ellipsis, DP-Ellipsis and E-type anaphora be analyzed as involving a syntactic process of covertly copying an item that functions as a variable bound by a quantificational antecedent. These 'copied bound variables' may take either the form of a bound proform or a Bound Trace ($[_D\ e_{x\sigma}\ ]$). Based upon the interpretations common to such 'copied bound variables' in VP-Ellipsis, DP-Ellipsis and E-type anaphora, we have also characterized them as more or less on a par with a demonstrative *that* (or *those*) in their meanings. (104a)–(106a) below present representative examples of each such construction, (104b)–(106b) present their LF-representations after bound variables are covertly copied, and (104c)–(106c) present paraphrases of 'copied bound variables' with a demonstrative expression.

(104) a. In our school, **every student**$_x$ [$_{VP}$ respects **his**$_x$ teacher]$_y$, and the parents also expect me to **[$_{VP}$ e]**$_y$.
b. LF: In our school, every student$_x$ [$_{VP}$ respects his$_x$ teacher]$_{y,}$ and
the parents also expect me to [$_{VP}$ respect [$_{DP}$ [$_{DP}$ **his**$_{x\sigma}$] **teacher**]]$_y$
c. In our school, every student$_x$ [$_{VP}$ respects his$_x$ teacher$_z$]$_{,}$ and
the parents also expect me to respect **that teacher**$_z$.

(105) a. *siritu-daigaku-no* **dono kyoozyu**$_x$*-ga* [$_{DP}$ **zibun**$_x$*-no gakusee*]*-o*
private-college-GEN which professor-NOM [$_{DP}$ self-GEN student]-ACC
suisensite-**mo**, *Monbusyoo-wa* **[$_{DP}$ e]** *saiyoosi-nai-daroo.*
recommend-even Ministry.of.Education-TOP employ-not-perhaps
'No matter which professor of a private college may recommend self's (= his or her own) student, the Ministry of Education will probably not employ those students.'
b. LF: ... *Monbusyoo-wa* **[$_{DP}$ [$_{DP}$ zibun$_{x\sigma}$]-no gakusee]**$_y$ *saiyoosi-nai-daroo*
... Min.of.Edu-TOP self-GEN student employ-not-perhaps
c. ... *Monbusyoo-wa* **sono gakusee**$_y$*-o* *saiyoosi-nai-daroo.*
... Ministry.of.Education-TOP **that student**-ACC employ-not-perhaps

(106) a. **Few congressmen**$_y$ admire Kennedy. **They**$_y$ are very junior.
b. LF: Few$_x$ [$_{DP}$ [$_D$ e$_{x\sigma}$] congressmen]$_y$ admire Kennedy.
[$_{DP}$ [$_D$ e$_{x\sigma}$] congressmen]$_y$ are very junior
c. Few congressmen$_y$ admire Kennedy.
Those congressmen$_y$ are very junior.

When we extend the same copying approach to the typical examples of donkey anaphora with ∀-readings and ∃-readings, we will obtain similar set of representations as in (107a–c) and (108a–c) below, respectively.

(107) a. Every farmer who owns **a donkey$_x$** beats **it$_x$**.
b. LF: [Every farmer who a_X owns $[_{DP}$ $[_D$ $e_{x\sigma}$] donkey $]_y$] beats **$[_{DP}$ $[_D$ $e_{x\sigma}$] donkey $]_y$**
c. Every farmer who owns a donkey beats **that donkey**.

(108) a. Every person who has a dime will put **it** in the meter.
b. LF: [Every person who a_X has $[_{DP}$ $[_D$ $e_{x\sigma}$] dime $]_y$] put **$[_{DP}$ $[_D$ $e_{x\sigma}$] dime $]_y$** in the meter
c. Every person who has a dime will put **that dime** in the meter.

Interestingly, the paraphrase with a demonstrative in (107c) can still exhibit ∀-readings and that in (108c) can still exhibit ∃-readings. This state of affairs is compatible with our approach, in which all instances of successful donkey anaphora are derived uniformly with the covert copying of a 'Bound Trace,' and the 'copied bound variable' has been observed to function like a demonstrative expression. The asymmetrical availability of ∃- or ∀-readings, in other words, is regarded in this approach as arising mainly from pragmatic factors, rather than a syntactic or semantic constraint.

6. Summary

In this work, I first pointed out that some ellipsis constructions exhibit a type of strict identity involving a bound proform, and argued that such an interpretation be derived by the covert copying of a bound proform. I then pointed out quite pervasive parallelism between such ellipsis constructions and E-type anaphora, and proposed to extend the covert copying approach from ellipsis to E-type anaphora. The proposed analysis crucially postulates the covert copying of a Bound Trace of the form $[_{DP}\ [_D\ e\ ]_{x\sigma}\ NP\ ]$, and the economy restriction imposed on the copied operators by the Minimal Variable Binding. It was also argued that the proposed approach can provide straightforward accounts for certain puzzles concerning agreement and quantifier scope when it is supplemented by the number system which distinguishes the overt number of the lexical form of a quantificational expression from the logical number of its Bound Trace. Finally, I have also explored the extension of the copying approach to donkey anaphora, while pointing out problems of some alternative analyses. It seems that the observations and generalizations we have presented in this work point toward the conclusion that when an item that functions as a variable bound by a quantificational expression is covertly copied at LF, it comes to function something akin to an anaphorically used demonstrative *that* (or *those*). Why this state of affairs obtains is certainly the question we must attempt to answer next, exploring its possible correlation to demonstrative expressions themselves. (See the last paragraph in Footnote 14 for a relevant observation.)

Notes

* An earlier and shorter version of this work appeared as Kitagawa (2000). I am extremely grateful to Hajime Hoji, Nam-kil Kim and Audrey Li for providing me with an opportunity to further develop the materials I had. Since I submitted this work to these editors during the

summer of 1999, I have developed a new approach to some of the problems taken up in this work. For various (mainly practical) reasons, however, I have decided to have this work published virtually in its original form only with the addition of my responses to the reviewers' comments. Part of the new approach was presented in Kitagawa (1999a), and I plan to further elaborate on it in my future work. I would also like to express my gratitude to Yasuaki Abe, Yoshio Endo, Tom Ernst, Steven Franks, Leslie Gabriele, Hajime Hoji, Kyle Johnson, Hideki Kishimoto, Roger Martin, Alan Munn, Mamoru Saito, Yukinori Takubo, Chris Tancredi, Satoshi Tomioka, Ayumi Ueyama, and Barbara Vance for their invaluable comments and judgments. I am especially grateful to Leslie Gabriele for her advice concerning the semantic aspects of the work, and Steven Franks for his criticism and encouragement on the extension of the minimalist hypothesis. Thanks are also due to all the participants of my syntax seminars at Indiana University in spring 1998 and spring 2001. Part of this work was also presented at Kobe University, Kyushu University, Nanzan University and University of Massachusetts at Amherst. This work is in part supported by the COAS Faculty Research Grant from Indiana University.

1 See Kitagawa (1991b), Kitagawa (1994), Kitagawa (1995), Kennedy (1997) and Fox (2000) for the motivation for such covert syntactic operations.

2 See Kitagawa (1991b) for the motivations to adopt this particular approach, and also for arguments that sloppy identity is independent of λ-abstraction. Another possible derivation for strict identity is to reconstruct an unbound proform *his* and let it be coreferential with the antecedent *John* in the first clause. See Kitagawa (1991b) for an analysis which does not leave this as a possibility.

3 In this work, I will tentatively label the ellipsis construction in question as DP-Ellipsis since it will allow us to reduce all the anaphoric properties we will deal with below to the category D(eterminer). It certainly requires further scrutiny to determine what categorial status nominal expressions in Japanese, especially those without phonetic content, have.

4 As illustrated by the interpretive restriction in (i) below, *zibun* must be bound.

(i) [$_2$ **Kebin Kosunaa**-*ga* *turetekita* *bodiigaado*]-*ga* **zibun**$_{2/*1}$-*no* *okusan-to* *odotta.*
Kevin Costner$_1$-NOM brought bodyguard-NOM self$_{2/*1}$-GEN wife-with danced
'The bodyguard Kevin Costner brought danced with **self**'s (= his own) wife.'

Strict identity in (6), in other words, cannot be derived in the alternative way described in Footnote 2.

Otani and Whitman (1991), following Huang (1991), argue that the sloppy identity observed in (4) is due to the hidden existence of VP-Ellipsis in Japanese. Many serious roblems of this 'Disguised VP-Ellipsis' approach have been pointed out, however, by Hoji (1998), Kim (1999) and Kitagawa (1999b).

5 Tomioka (1998) presents the examples in (i) below and points out that sloppy identity for pronouns of laziness in English have tighter restriction than that for the empty proforms in Japanese as in (4).

(i) a. Gary likes **his mother**. Tim likes **her**, too.
– *her* = *Tim's mother
b. Gary lost **his ID** in the gym. Tim lost **it** in a classroom.
– *it* = ???Tim's ID
c. Gary thinks **his teachers** are geniuses, but Tim thinks **they** are nuts.
– *they* = ???Tim's teachers

With somewhat tighter sequencing of eventualities, and especially with genericity, as in (ii) below, however, sloppy identity becomes possible in similar discourses.

(ii) a. Quite often, a young husband has not learned the proper way of expressing his affection to **his wife**, but an old man usually knows how to please **her.**
– *her* = an old man's wife

b. John and Bill have totally different policies concerning the upbringing of their own children. While John disciplines **his children** quite often, Bill tries to let **them** learn right from wrong on their own.
– *them* = Bill's children

c. Many dog lovers walk **their dogs** in the park, but of course there are thousands of people who have no choice but to walk **them** on the street.
– *them* = the dogs of thousands of people.

Kitagawa (1999a) attempts to account for both these phenomena, making an appeal to the notion 'economy of lexical information' and postulating a purely syntactic version of Hoji's (1998) 'supplied N Head' analysis (See the analysis in (29) below) in addition to DP-Copy and VP-Copy. In that work, it is also pointed out that the same approach allows us to explain why Japanese exhibits DP-Ellipsis while English exhibits VP-Ellipsis to represent similar sloppy identity interpretations.

6 The singular (and distributive) reading 'that teacher' is obtained in (15) when *the parents* is interpreted as the parents of each student, and the plural (and group) reading 'those teachers' is obtained when *the parents* is interpreted as the parents of the students as a whole. Perhaps, grammar also permits sloppy identity here, although we will concentrate on strict identity in our discussion.

7 The pronoun *him* in (i) below cannot strictly refer to *Bill.*

(i) Mary blamed **him**$_{*1}$, and **Bill**$_1$ did $[_{VP}$ e], too.

As has been discussed in Kitagawa (1991b: 504–505), this puzzling interpretive restriction can be explained if the elided VP is reconstructed in covert syntax, and the resulting LF-representation is subject to the Condition B of the Binding Theory, as illustrated in (ii).

(ii) LF: Mary blamed him$_1$, and **Bill**$_1$ did $[_{VP}$ blame **him**$_{*1}$], too

VP-Ellipsis involving *do*, in other words, does in fact exhibit properties that can be captured if it involves syntactic reconstruction and binding rather than pragmatic control.

8 I am grateful to Hajime Hoji and Ayumi Ueyama for useful discussion on this issue.

9 Fiengo and May (1994: 269) report that the strict identity in question is possible in the antecedent-contained deletion as in (i).

(i) **The men**$_1$ $[_{VP}$ introduced **each other**$_1$ to everyone that *the women* did $[_{VP}$ e]].

To my dismay, most of over ten speakers I have checked with reject such a reading, although they accept a sloppy identity interpretation. More importantly, all of them including the few who are not sure if strict identity is absolutely impossible in (i) find a similar sentence as in (ii) below ungrammatical.

(ii) **The men**$_1$ $[_{VP}$ introduced **each other**$_1$ to everyone that ***I*** { did $[_{VP}$ e] / wanted to $[_{VP}$ e] }].

Note that the plural subject *the women* in (i) as the potential local antecedent of the reconstructed anaphor is replaced by the singular subject *I* in (ii). The ungrammaticality of (ii) then would lead us to the rejection of the vehicle change analysis in both (i) and (ii). See

Kitagawa (1991b: 527), however, for some ideolectal variations concerning sloppy identity involving *each other*. Note also that the ungrammaticality of (24) would remain unaccounted for in the pragmatic account of the strict identity in (17). Kitagawa (1991a) offers an account of the contrast between (17) and (24).

10 See Huang (1983) and Kitagawa (1994: 355–357), among others, for the discussion on the constraint that anaphors must be bound by the closest antecedent.

11 Covert copying is assumed to be applicable within a single discourse, if necessary, in an 'interarboreal' fashion across sentence and utterance boundaries. This may be regarded as another significant departure of our analysis from Chomsky's narrowly defined 'minimalist program.' In an approach in which syntactic objects are generated 'bottom-up' rather than 'top-down,' such an analysis is not out of the question. Note also that PF deletion for ellipsis adopted by Chomsky must be applicable in a similar 'interarboreal' fashion.

12 Under Chomsky (1995) 'Copy plus Delete' analysis of movement, covert copy here can be regarded as a case of covert movement in which Copy applies without being followed by the application of Delete when neither the extraction site nor the landing site c-commands the other.

13 Depending on the analysis of quantificational determiners and possessors in the pre-nominal position, the two constructions may turn out to look even more parallel. See Abney (1987) for relevant discussion.

14 In the proposed approach, the Minimal Variable Binding in (57) does not rule out a sentence like (ia) below, whose semantics is often represented as in (ib), providing us with the impression that the operator can in fact establish more than one instance of operator-variable binding.

(i) a. Every student respects **his** teacher.
b. $\forall$x [student (x) $\rightarrow$ respect (x, **x's** teacher)]

Whichever derivation in (ii) below we may adopt in the proposed approach, however, *every* establishes operator-variable binding only once in the derivation with its trace $[_D\ e\ ]_x$, and the pronoun *his* establishes its variable status by being bound by $[_{DP}\ e\ ]_y$ in (iia) and by $[_{DP}\ [_D\ e\ ]_x$ *student* $]_y$ in (iib).

(ii) a. Every$_x$ $[_{DP}$ $[_D$ e $]_x$ student $]_y$ $[_{VP}$ $[_{DP}$ e $]_y$ respects **his**$_y$ teacher]
b. Every$_x$ $[_{DP}$ $[_D$ e $]_x$ student $]_y$ respects **his**$_y$ teacher

We may also be able to develop an alternative to the proposed economy account, making an appeal to the notion 'type-mismatch' which Heim and Kratzer (1998) regard as the trigger for the QR of object QPs. Note that the covert reconstruction involved in (54a) copies the QP *few congressmen* (type <<e, t>, t>) onto the pronoun *they* (type <e>). Either this mismatch may prohibit LF-copy itself or, it may rule out the resulting representation later. Then, in the legitimate case of LF-copy as in (49b), the bound D-variable ($[_D\ e\ ]_{x\sigma}$) as the specifier of the Bound Trace ($[_{DP}\ [_D\ e_{x\sigma}\ ]$ *congressmen* $]_y$) is to be analyzed as an entity of the type (<<e, t>, e>) since it maps an entity of the type <e, t> to that of a type <e>. This result is compatible with our intuition that a strict variable exhibits an interpretation akin to that of an anaphoric demonstrative like *that*, which can be also analyzed as an entity of the type <<e, t>, e> in a sentence like (iii).

(iii) I know $[_{DP}$ *that* $[_{NP}$ congressman]].

15 The use of *boy* in (59a–c) assures that *they* as a singular pronoun is not used in these examples simply to avoid the mention of gender as in (i).

(i) Everyone knows that { **they** / #**he** } should apologize.

16 The so-called telescoping poses an exception to this generalization. It is quite possible that telescoping involves the covert copying of an N-projection rather than a DP. Such an analysis is

compatible not only with the logical singularity a universal quantifier exhibits in telescoping but also with the fact that telescoping must be licensed by genericity in a broad sense. See Poesio and Zucchi (1992) and references therein for relevant discussion. Kitagawa (1999a) attempts to account for telescoping in the approach mentioned in Footnote 5 above.

17 cf. May's (1985) semantic number of quantificational elements.

18 Certain non-quantificational nominal expressions like *family*, *audience*, and *committee* may exhibit flexibility in their **overt** numbers since their singular forms can refer to either a unit or the members of the unit:

(i) a. the **family** that **has** just moved in
b. My **family are** all very well.

19 See, for instance, Fodor and Sag (1982), Heim (1982) and Chierchia (1995) for relevant discussion.

20 See, for example, May (1985) and Lappin (1988/89).

21 Providing examples like (i) and (ii) below, Lappin (1988/89) convincingly argues that the antecedent of a donkey pronoun (as a type of E-type pronoun) need not contain a weak D (contra Reinhart (1987)), and that there is no need for the quantifier containing the antecedent to c-command the donkey pronoun (contra Haïk (1984)). Note that in (i) the set of *at least half the films at the festival* need not be identical for each critic:

(i) Every critic who saw **at least half the films at the festival**$_x$ liked **them**$_x$.
(ii) John spoke to [every student who submitted **a paper**$_x$] about the possibility of publishing **it**$_x$.

22 The notion 'downward-entailing' operators does not seem to be relevant, either, since *at least n* and *many*, which are not downward entailing, can be (overtly as well as) logically plural. I am grateful to Leslie Gabriele for helping me clarify the relevant notion here.

23 The non-elliptical sentence here actually must be accompanied by proper prosodic reduction of the repeated I' (*will beat it*), and preferably by the presence of one of the 'same saying' operators *too* at the end of the sentence, to justify the repetition of the identical I'.

References

Abney, S. (1987). *The English Noun Phrase in Its Sentential Aspect*. Doctoral dissertation, Massachusetts Institute of Technology.

Chierchia, G. (1995). *Dynamics of Meaning: Anaphora, Presupposition, and the Theory of Grammar*. Chicago: The University of Chicago Press.

Chomsky, N. (1995). *The Minimalist Program*. Cambridge, Massachusetts: The MIT Press.

Fiengo, R. and R. May. (1994). *Indices and Identity*. Cambridge, Massachusetts: The MIT Press.

Fodor, J. and I. Sag. (1982). Referential and Quantificational Indefinites. *Linguistics and Philosophy* **5**, 355–398.

Fox, D. (2000). *Economy and Semantic Interpretation*. Cambridge, Massachusetts: The MIT Press.

Haïk, I. (1984). Indirect Binding. *Linguistic Inquiry* **15**, 185–223.

Hankamer, J. and I. Sag. (1976). Deep and Surface Anaphora. *Linguistic Inquiry* **7**, 391–428.

Heim, I. (1982). *Semantics of Definite and Indefinite Noun Phrases*. Doctoral dissertation, University of Massachusetts, Amherst.

Heim, I. and A. Kratzer. (1998). *Semantics in Generative Grammar*. Oxford: Blackwell Publishers, Inc.

Higginbotham, J. (1983). Logical Form, Binding and Nominals. *Linguistic Inquiry* **14**, 395–420.

Hoji, H. (1998). Null Object and Sloppy Identity in Japanese. *Linguistic Inquiry* **29**, 127–152.

Huang, C. T. J. (1983). A Note on the Binding Theory. *Linguistic Inquiry* **14**, 554–561.

Huang, J. (1991). Remarks on the Status of the Null Object. In *Principles and Paremeters in Comparative Grammar*, edited by Freidin, Robert, 56–76. Cambridge: MIT Press.

Jacobson, P. I. (1980). *The Syntax of Crossing Coreference Sentences*. New York: Garland Publishing, Inc.

Kennedy, C. (1997). Antecedent-Contained Deletion and the Syntax of Quantification. *Linguistic Inquiry* **28**, 662–688.

Kim, S. (1999). Sloppy/Strict Identity, Empty Objects, and NP Ellipsis. *Journal of Asian Linguistics* **8**, 255–284.

Kitagawa, Y. (1991a). Binding under the Internal Subject Hypothesis. In *Trends in Linguistics: State of the Art Report 16*, edited by Nakajima, Heizo, 205–259. Mouton de Gruyter.

Kitagawa, Y. (1991b). Copying Identity. *Natural Language & Linguistic Theory* **9**, 497–536.

Kitagawa, Y. (1994). *Subjects in Japanese and English*. New York: Garland.

Kitagawa, Y. (1995). 'Affect' in Binding and Coreference. In *FLSM VI, Proceedings of the Sixth Annual Meeting of the Formal Linguistics Society of Mid-America*, 130–142. Indiana University Linguistics Club.

Kitagawa, Y. (1999a). Economy of Lexical Selection. Paper presented at LICSSOL 1 Economy in Language Design, Computation and Use, The Lyon Institute for Cognitive Science,

Kitagawa, Y. (1999b). VP-Ellipsis vs. NP-Ellipsis. In *Linguistics: In Search of Human Mind A Festschrift for Kazuko Inoue*, edited by Muraki, Masatake and Enoku Iwamoto, 347–372. Tokyo: Kaitakusha.

Kitagawa, Y. (2000). Ellipsis and E-type Anaphora. In *University of Massachusetts Occasional Papers in Linguistics 24*, edited by GLSA.

Lappin, S. (1988/89). Donkey Pronouns Unbound. *Theoretical Linguistics* **15**, 263–286.

May, R. (1985). *Logical Form: Its Structure and Derivation*. Cambridge, Massachusetts: The MIT Press.

Otani, K. and J. Whitman. (1991). V-Raising and VP-Ellipsis. *Linguistic Inquiry* **22**, 345–358.

Poesio, M. and A. Zucchi. (1992). On Telescoping. *Proceedings of the Second Conference on Semantics and Linguistic Theory* 347–366.

Reinhart, T. (1987). Specifier and Operator Binding. In *The Representation of (In)definiteness*, edited by Reuland, E.J. and A. G. B. ter Meulen, 130–67. Cambridge: MIT Press.

Tomioka, S. (1998). *The Semantics of Japanese Null Pronouns*. Manuscript, Universität Tübingen.

Williams, E. (1977). Discourse and Logical Form. *Linguistic Inquiry* **8**, 101–139.